

CN-NEWTTS BENCH: A TARGET-LEVEL AUTOMATIC BENCHMARK FOR RAW-INPUT CHINESE NEWS TTS PRONUNCIATION

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ABSTRACT

Chinese news text contains dense written forms such as scores, hyphenated model names, ranges, unit symbols, percentages, English abbreviations, and mixed Chinese–Latin–digit names. These forms are frequent in real listening workflows, and a text-to-speech (TTS) system can preserve the written string while changing the spoken meaning. We introduce CN-NewsTTS Bench v0.1, an open target-level benchmark for evaluating whether Chinese news TTS products pronounce such targets correctly from raw text, without user-side rules, LLM rewriting, SSML hints, or manual edits. The release contains a 200-record development set, an 800-record public test set, 992 public auto-evaluable targets, fixed transcripts from a three-ASR ensemble, an automatic target scorer, and initial results for seven product TTS systems. We additionally report ASR-route diagnostics, ASR-subset ablations, category-level results, confidence intervals, and provider configuration metadata. The best system reaches 0.879 strict accuracy, while several systems remain below 0.60.

Index Terms— text-to-speech, Chinese news, pronunciation evaluation, text normalization, ASR-based evaluation

1. INTRODUCTION

Modern Chinese news TTS systems are used in settings where listeners expect symbols and compact written forms to be read according to news conventions. A short article may contain a sports score such as 96-91, an aircraft model such as 苏-27, a year range such as 2028-2030年, a torque unit such as 620N·m, or a generation label such as 80后. These strings are usually unambiguous to a human editor, yet the same surface marks can represent ranges, subtraction, model designators, unit names, or ordinary numbers.

Such failures are not merely naturalness problems. Reading a score as a range changes the relation between the numbers; reading 苏-27 as a negative number changes a military model into a nonsensical phrase; reading 80后 as 八十后 loses the conventional generation-label reading. Subjective tests such as MOS remain important for speech quality [1], and objective predictors can estimate global naturalness or voice-conversion quality [2]. They are, however, too coarse to track specific high-impact reading decisions. Generic ASR round-trip comparison is also insufficient because the evaluation must decide whether a particular target was read as an intended news form.

The problem is closely related to text normalization and non-standard-word handling in TTS [3, 4, 5, 6], Mandarin text normalization [7, 8], and Chinese G2P/polyphone benchmarks [9, 10]. CN-NewsTTS Bench differs by measuring the end-to-end raw-input behavior exposed by deployed TTS APIs rather than an isolated frontend component. The main contributions are:

- an open, deterministic dev/public split for Chinese news pronunciation targets;
- a Raw Input Product Track that excludes user-side normalization or hints;
- a target-level schema with positive and negative readings;
- a three-ASR automatic scoring protocol with route diagnostics and ablations; and
- a reproducible v0.1 leaderboard for seven product TTS systems.

2. BENCHMARK DESIGN

2.1. Raw Input Product Track

The main track measures product-facing behavior under a fixed raw-input condition. Each system receives the same original Chinese news-style text. Provider-internal text normalization and frontend behavior are allowed, but external rule frontends, LLM rewrites, SSML pronunciation hints, and manual benchmark-text edits are not allowed. A fixed voice is chosen before evaluation and kept constant for all samples. This is therefore a raw-input product evaluation, not a claim that the benchmark isolates model weights or removes all provider-side frontend logic.

2.2. Data Construction and Targets

The v0.1 data are deterministic synthetic news-style sentences generated from category-specific templates and lexicons, not reused copyrighted news articles. The released generator script uses seed 20260620, 11 template families, and 86 lexicon entries covering teams, events, projects, cities, companies, military models, vehicle models, abbreviations, mixed brand strings, unit symbols, generation labels, and polyphone audit items. The templates mirror production risk motifs such as sports recaps, defense/vehicle models, technical units, mixed brands, and time spans; they are diagnostic rather than a claim about full news frequency. The dev and public splits are constructed from fixed per-category quotas, then shuffled. Each record contains targets with offsets, categories, positive/negative readings, and an auto-evaluable flag. Optional same-character polyphones are included for audit but excluded from the main automatic score because ordinary ASR text often hides tonal distinctions.

Table 1. CN-NewsTTS Bench v0.1 data scale.

Split	Rec.	Targets	Auto	Opt.
dev	200	252	248	4
public	800	1008	992	16
total	1000	1260	1240	20

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Table 2. Representative target annotations. Positive and negative patterns are simplified examples; the released JSONL contains the full pattern lists.

Span	Type	Positive	Negative
96-91	score	九十六比九十一	九十六到九十一
苏-27	model	苏二七; 苏二十七	苏负二十七
80后	generation	八零后	八十后
620N·m	unit	六百二十牛米	六百二十恩点米

The public split contains 224 sports-score targets, 192 military-model targets, 104 abbreviations, 96 unit-symbol targets, 80 hyphenated ranges, 80 year ranges, 64 mixed brand names, 64 vehicle models, 56 percentages, and 32 generation labels. The public test set is a transparent diagnostic split rather than a long-term anti-overfitting competition set; a hidden split can be added if leaderboard gaming becomes a practical concern.

3. AUTOMATIC EVALUATION

3.1. Three-ASR Protocol

The v0.1 public leaderboard uses three heterogeneous ASR routes: MiMo API ASR, SenseVoiceSmall, and Paraformer-zh. SenseVoiceSmall and Paraformer-zh are open-source local recognizers from the FunAudioLLM/FunASR ecosystem [11, 12]; MiMo API ASR provides an API-based route. The public release fixes all three ASR transcript files, so the leaderboard can be reproduced without rerunning commercial TTS APIs or ASR services. For seven TTS systems and 800 public records, the canonical transcript set contains 16,800 successful ASR rows and no missing public samples.

3.2. Target-Level Voting and Metrics

For each ASR transcript and target, the scorer applies Unicode, case, punctuation, and whitespace normalization. It then matches negative-reading patterns before positive-reading patterns. A single ASR route returns `wrong` if a negative reading is found, `correct` if a positive reading is found, and `unknown` otherwise. Majority voting gives final `correct` or `wrong` if at least two routes agree; all remaining combinations become `unknown`.

The primary metric is strict auto accuracy,

$$\text{StrictAcc} = \frac{\# \text{correct}}{\# \text{auto-evaluable targets}}. \quad (1)$$

We also report coverage and resolved accuracy:

$$\text{Coverage} = \frac{\# \text{correct} + \# \text{wrong}}{\# \text{auto-evaluable targets}}, \quad (2)$$

$$\text{ResolvedAcc} = \frac{\# \text{correct}}{\# \text{correct} + \# \text{wrong}}. \quad (3)$$

Keeping unknowns in the strict denominator prevents a system from appearing accurate only on the subset that the ASR ensemble can resolve.

3.3. ASR Protocol Diagnostics

Table 3 pools all 6,944 public target instances (7 systems $\times$ 992 targets). The three individual routes have similar strict accuracy but different coverage behavior. MiMo API ASR is conservative: it has high resolved accuracy when it resolves a target, but many unknowns.

Table 3. ASR-route diagnostics over 6,944 public target instances. R-Strict denotes strict accuracy computed from one ASR route before majority voting.

Route	R-Strict	Cov.	Res.	Unk.
MiMo API ASR	.513	.569	.902	.431
Paraformer-zh	.543	.748	.725	.252
SenseVoiceSmall	.537	.728	.738	.272
Majority vote	.540	.730	.739	.270

Paraformer-zh and SenseVoiceSmall resolve more targets. Pairwise tri-label agreement is 0.959 between Paraformer-zh and SenseVoiceSmall, but 0.595 and 0.604 between MiMo API ASR and the two local routes, showing that the ensemble is heterogeneous. Overall, 41.7% of target instances have at least one route disagreement, while only 0.7% are complete correct/wrong/unknown ties.

Table 4. ASR-subset ablation for strict accuracy. P, S, and M denote Paraformer-zh, SenseVoiceSmall, and MiMo API ASR; two-route settings require agreement.

System	Full3	P+S	M+S	M+P
Volcano	.879	.870	.575	.570
Azure	.756	.750	.504	.510
Google	.604	.592	.404	.408
MiniMax	.548	.542	.366	.360
Aliyun	.472	.468	.337	.339
MiMo	.275	.274	.225	.226
AWS	.244	.230	.192	.200

The ASR ablation in Table 4 preserves the complete seven-system rank order under all three two-route subsets. The local Paraformer+SenseVoice route changes only 88 of 6,944 final target labels (1.27%) relative to the three-route majority, and all such changes turn resolved labels into `unknown`. We use the three-route majority because the two local routes are highly correlated, while MiMo API ASR adds an independent route. The lower M+S and M+P scores are a two-route coverage effect: both routes must resolve and agree, so conservative MiMo outputs become `unknown`. Thus the two-route variants are stability checks, not the leaderboard protocol.

4. INITIAL PUBLIC RESULTS

We evaluate seven product TTS systems: Volcano/Doubao TTS, Azure Speech TTS, Google Cloud TTS, MiniMax TTS, Aliyun CosyVoice, MiMo TTS, and AWS Polly. Each system generates one audio file per public test record, using one fixed voice and provider-level default normalization. Generated audio is not redistributed in the GitHub repository because commercial provider terms differ; fixed ASR transcripts are released as the reproducible scoring artifact.

Figure 1 and Table 5 show substantial variation. The best system reaches 0.879 strict accuracy, but several widely used systems remain below 0.60. Azure has no majority-voted wrong targets, but it has 242 unknown targets, so its strict accuracy is 0.756 rather than 1.0. Google and MiniMax have higher coverage, but many more wrong targets. These patterns show why strict accuracy, coverage, and resolved accuracy must be reported together.

Table 6 shows that the benchmark is not a single global difficulty score. Percentages, vehicle models, and abbreviations are largely solved by the evaluated systems. Sports scores are the hardest category by strict accuracy and wrong rate: many systems read

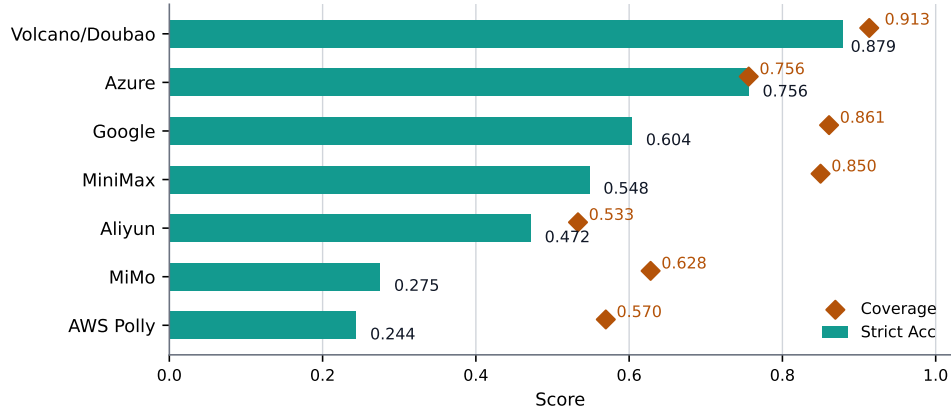


Fig. 1. Public test strict accuracy and coverage for seven raw-input TTS products.

Table 5. Public test results on 992 auto-evaluable targets. Confidence intervals are Wilson 95% intervals for strict accuracy.

System	Strict	95% CI	Cov.	Res.	C/W/U
Volcano	.879	[.857,.898]	.913	.962	872/34/86
Azure	.756	[.728,.782]	.756	1.000	750/0/242
Google	.604	[.573,.634]	.861	.701	599/255/138
MiniMax	.548	[.517,.579]	.850	.645	544/299/149
Aliyun	.472	[.441,.503]	.533	.885	468/61/463
MiMo	.275	[.248,.304]	.628	.438	273/350/369
AWS	.244	[.218,.272]	.570	.428	242/323/427

score hyphens as ranges or subtractions. Unit symbols have the highest unknown rate, often because ASR routes cannot reliably expose symbol-level readings.

The category averages hide system-specific failure modes, so Table 7 breaks out the hardest categories by system.

Table 7. Per-system strict accuracy on selected hard categories.

System	Sports	Unit	Mil.	Brand	Gen.
Volcano	.996	.573	.865	.391	1.000
Azure	.504	.646	.771	.438	1.000
Google	.000	.417	.776	.484	.125
MiniMax	.000	.542	.448	.484	.562
Aliyun	.134	.021	.198	.281	.750
MiMo	.000	.198	.000	.438	.000
AWS	.000	.000	.026	.438	.125

Volcano nearly solves sports scores but still leaves units and mixed brands difficult; Azure is conservative but strong on generation labels and units; Google and MiniMax lose most sports-score targets even when they are competitive elsewhere. For zero-score sports systems, the dominant failure is reading score hyphens as ranges, e.g., 96-91 as 九十六到九十一. Route-level range negatives dominate subtraction negatives (1,496 vs. 266).

5. REPRODUCIBILITY AND RELEASE

Repository: <https://github.com/Jayden-X-L/cn-news-tts-bench>

Release: <https://github.com/Jayden-X-L/cn-news-tts-bench/releases/tag/v0.1>

Table 6. Category-level public results pooled over the seven TTS systems. The target count is per system; rates are over all systems.

Category	N	Strict	Cov.	Wrong	Unk.
percentage	56	1.000	1.000	.000	.000
vehicle	64	.929	.951	.022	.049
abbrev.	104	.865	.912	.047	.088
range	80	.750	.827	.077	.173
year range	80	.713	.713	.000	.288
generation	32	.509	.813	.304	.188
military	192	.440	.587	.147	.413
brand	64	.422	.578	.156	.422
unit	96	.342	.528	.186	.472
sports	224	.233	.728	.494	.272

Commit: `f94a679fc7fc`. The release includes data, schema, scoring code, fixed ASR transcripts, leaderboard files, the dashboard, audit, and checksums. Table 8 lists compact model and voice labels; region, sample rate, and invocation date are recorded in the release metadata. The v0.1 data, TTS, and audit dates are 2026-06-20, 2026-06-22, and 2026-06-23.

Table 8. Initial TTS configuration.

System	Model/API	Voice/config label
Volcano	seed-tts-2.0-standard	default zh
Azure	azure-neural-tts	zh-CN-XiaoxiaoNeural
Google	Chirp 3 HD	cmn-CN-Chirp3-HD-Kore
MiniMax	speech-2.8-hd	Mandarin news
Aliyun	cosyvoice-v3-plus	longanyang
MiMo	mimo-v2.5-tts	白桦
AWS	polly-neural	Zhiyu

A minimal reproduction consists of validating the public split, scoring a per-model ASR result file, aggregating the leaderboard, and verifying the checksum manifest. The three canonical ASR transcript files each contain 5,600 rows. New TTS systems can be evaluated by generating audio from the raw public text and either running the provided ASR pipeline or submitting ASR result files in the documented JSONL format.

Leaderboard entries should report strict accuracy, coverage, category breakdowns, data release, provider, voice, region, sample rate, date, and ASR files. Updated APIs should be added as dated entries rather than replacing the baseline.

6. LIMITATIONS

The benchmark is automatic and should not replace human listening tests. ASR text can hide pronunciation errors, especially same-character tones, and unknown rates vary by category. The public split supports reproducibility but may later require a hidden test split. Commercial TTS audio is omitted because provider terms vary; the release fixes data, metadata, transcripts, scores, and checksums for dated comparison.

7. CONCLUSION

CN-NewsTTS Bench v0.1 provides an open, target-level benchmark for raw-input Chinese news TTS pronunciation accuracy. By fixing the data split, three-ASR transcripts, scorer, category diagnostics, and initial seven-system leaderboard, it makes a common production failure mode measurable and reproducible. The results indicate that compact Chinese news forms remain a practical challenge for current TTS products.

8. COMPLIANCE WITH ETHICAL STANDARDS

This work uses synthetic news-style text and released benchmark transcripts. No human-subject audio, personal speech data, or private user data is released.

9. REFERENCES

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